

6.2

$$V = \text{span} \{ (2, 1, 0, 1)^T, (1, 0, 1, 2)^T \} \leq \mathbb{R}^4$$

$$f((2, 1, 0, 1)^T) = (4, 1, 2, 5)^T$$

$$f((1, 0, 1, 2)^T) = (2, 2, 2, 2)^T$$

$$\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} + b \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} \text{ jsou LN.} \rightarrow B = \left(\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right) \text{ je báze } V$$

$$\begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix} = a \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + b \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \quad \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \rightarrow [f]_B^B = \begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2 \\ 2 \\ 2 \\ 2 \end{pmatrix} = c \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + d \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \quad \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} \rightarrow$$

$$[f]_B^B - \lambda I_2 = \begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 1-\lambda & 2 \\ 2 & 2-\lambda \end{pmatrix}$$

$$\det [f]_B^B = (1-\lambda) \cdot (2-\lambda) - 2 \cdot 2 = -2 - \lambda + 2\lambda + \lambda^2 - 4 = \lambda^2 + \lambda - 6 = (\lambda - 2) \cdot (\lambda + 3)$$

$$\lambda = 2 \text{ a } \lambda = -3$$

$$[M_2]_B = [\text{Ker}(f - \text{id}_V)]_B = \text{Ker}([f]_B^B - 2I_2) = \text{Ker} \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}$$

$$M_2 = [M_2]_K = \text{span} \left\{ [\text{id}]_K^B \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\} = \text{span} \left\{ 2 \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} + 1 \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\} = \text{span} \left\{ \begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix} \right\}$$

$$[M_{-3}]_B = [\text{Ker}(f - \text{id}_V)]_B = \text{Ker}([f]_B^B - (-3)I_2) = \text{Ker} \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right\}$$

$$M_{(-3)} = \text{span} \left\{ 1 \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} - 2 \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\} = \text{span} \left\{ \begin{pmatrix} -3 \\ -2 \\ 1 \\ 0 \end{pmatrix} \right\}$$

Nyní vezme vektor z M_2 a vektor z $M_{(-3)}$ a vytvoříme bázi C prostoru V . Vektory patří do V a jsou LN, jelikož oba patří jinému vlastnímmu číslo.

$$C = \left(\begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -2 \\ -3 \end{pmatrix} \right) \quad \text{a vypočítáme matici } f \text{ vzhledem k bázi } C: \quad [f]_C^C = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}.$$

$$\text{Jak bude vypadat } f^n? \quad [f^n]_C^C = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}^n = \begin{pmatrix} 2^n & 0 \\ 0 & (-3)^n \end{pmatrix}$$

Převědeme vzhledem ke kanonické bázi:

$$[(f^n)]_K^K = [\text{id}]_K^C \cdot [f^n]_C^C \cdot [\text{id}]_C^K$$

$$[\text{id}]_K^C = \begin{pmatrix} 5 & 0 \\ 2 & 1 \\ 1 & -2 \\ 4 & -3 \end{pmatrix}$$

$$([\text{id}]_K^C)^{-1} \rightarrow \begin{pmatrix} 5 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 5 & -2 & 5 \end{pmatrix} \sim \begin{pmatrix} 5 & 0 & 1 & 0 \\ 0 & 5 & -2 & 5 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & -\frac{1}{5} & 1 \end{pmatrix}$$

$$([\text{id}]_C^K)^{-1} = \begin{pmatrix} \frac{1}{5} & 0 & 0 & 0 \\ -\frac{1}{5} & 1 & 0 & 0 \end{pmatrix}$$

$$[(f^n)]_K^K = \begin{pmatrix} 5 & 0 \\ 2 & 1 \\ 1 & -2 \\ 4 & -3 \end{pmatrix} \cdot \begin{pmatrix} 2^n & 0 \\ 0 & (-3)^n \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{5} & 0 & 0 & 0 \\ -\frac{1}{5} & 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 5 \cdot 2^n & 0 \\ 2 \cdot 2^n & (-3)^n \\ 2^n & -2(-3)^n \\ 4 \cdot 2^n & -3(-3)^n \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{5} & 0 & 0 & 0 \\ -\frac{1}{5} & 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 2^n & 0 & 0 & 0 \\ \frac{2}{5} 2^n - \frac{2}{5} (-3)^n & (-3)^n & 0 & 0 \\ \frac{1}{5} 2^n + \frac{1}{5} (-3)^n & -2(-3)^n & 0 & 0 \\ \frac{4}{5} 2^n + \frac{1}{5} (-3)^n & -3(-3)^n & 0 & 0 \end{pmatrix}$$

$$f^n((5, 1, 3, 7)^T) = [(f^n)]_K^K \cdot \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} = \begin{pmatrix} 5 \cdot 2^n \\ 2^{n+1} - (-3)^n \\ 2^n - (-3)^n \\ 2^{n+2} + 3(-3)^n \end{pmatrix}$$